

5ai Random means it is impossible to state exactly which nucleus or when a particular nucleus will disintegrate, but only the probability of decay in unit time interval for a particular radioactive isotope. [B1]

a ii $\Delta N = -\lambda N \Delta t = 4.8 \times 10^{-11} \times 6.9 \times 10^{14} \times (5 \times 3.2 \times 10^7)$
 $= 5.2 \times 10^{12}$

a iii As the half life of americium is 458 years, N will not change much (remain constant) over 5 year period. N/t will remain constant over 5 years, thus this equation can be used.

OR

As the half life of americium is 458 years, over a few hundred years, N will decrease significantly. Since activity depends on number of nuclei present, N/t will also change significantly over a few hundred years. Thus this equation cannot be used.

a iv

Over the five years, dust and other particles might have built up on the surface of the detector. This will affect the detector's ability to detect the incoming alpha particles which will trigger a false alarm. Thus a replacement must be done.

OR

Alpha particles have strong ionising power. Over the 5 years, the electronic component in the detector will be affected by this high ionising power, leading to electronic failure inside the detector. Thus a replacement must be done.

b

Beta particles have lower penetration (have a shorter range or less likely to escape from the body) are more highly ionising than gamma photons.

The body or cells will have a larger (absorbed) dose of beta particles which will increase the risk of Cancer (or higher of mutation of cells)

ACCEPT reverse argument for gamma photons

6(a)(i) Force = $qE = 1.4 \times 10^{-15} (3.0 \times 10^3) = 4.2 \times 10^{-12} \text{ N}$